



A comparison study on optical character recognition models in mathematical equations and in any language

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ABSTRACT

Optical Character Recognition[OCR] is a technology that makes use of artificial intelligence and machine learning to extract readable text from documents, images, tags or any other type of sources. It allows one to convert characters and text objects into digital data that can be easily processed, analyzed, and modified. OCR can be applied to various types of languages in both written and spoken format. It can process everything from hand-written documents to typed-out text, making it a highly versatile technology. OCR makes use of a variety of algorithms and methods to process images, and then produces readable output, whatever language it is used for. This technology has the potential to be used for industries, banking, the medical field, security, and document storage among others. OCR faces significant challenges in accurately predicting language and mathematical expressions due to variations in handwriting styles, complex layouts, and the ambiguity of symbols. In this research, we propose assessing the results of different models that have been trained to identify an improved OCR system. The best OCR model is With the help of a decision tree model chosen.

1. Introduction

A short while ago, machine learning gained prominence in the area of experimentation attributable to its proven success in solving real-world problems. Machine learning is a field within artificial intelligence that enables systems to learn from data and improve their performance over time without the need for detailed programming for each specific task. The potential applications of machine learning are vast, ranging from autonomous driving systems to medical diagnosis and predicting consumer behavior. As a result, the field has attracted considerable interest from researchers and industry practitioners alike [1].

Optical Character Recognition (OCR) is an critical field that involves the recognition of printed or handwritten characters from images or documents [2]. Optical Character Recognition software uses algorithms to recognize and extract the characters in scanned images and convert them into machine-readable text. This technology is commonly used for digitalizing printed documents, such as books and financial records, and converting them into searchable and editable formats. OCR is often incorporated into document management systems, digitalization projects, and digital entry applications. OCR can also be used to recognize handwriting, although this application is less commonly used due to greater variability in handwriting samples. Traditional OCR systems were based on rule-based algorithms, which required a significant amount of human intervention and limited accuracy. The traditional machine learning models include Support Vector Machines(SVM), Naive Bayes, and k-nearest Neighbors(k-NN). In recent years, the development of machine learning models has provided significant improvements in OCR performance [3].

This paper aims to investigate the use of different machine learning models in Optical Character Recognition (OCR). Investigation of the performance of deep learning models, such as Convolutional Neural Networks(CNNs), Recurrent Neural Networks(RNNs), and

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their variations is performed. Standard OCR datasets, such as MNIST and IAM, are used to analyze the different machine learning models [4]. This study intends to investigate the implementation of ML approaches in solving an important real-world problem, to create an OCR system that can be implemented using Tesseract, EasyOCR, Paddle OCR, MMOCR, Keras OCR, i2OCR, and compare the most accurate model for each language. We prepared a dataset of 52 mobile phone photos that represent a wide range of real-world situations to thoroughly test optical character recognition (OCR) systems. We evaluated the models' performance using a range of metrics, including accuracy, precision, recall, and F1-score. The best OCR model is chosen with the help of a decision tree model. The problem requires analyzing large datasets and making accurate predictions, which can be demanding due to the complexity and multi-dimensional nature of the data. Overall, this paper aims to contribute to the advancement of machine learning methods in solving real-world problems and improving their performance when applied to large datasets [5]. Overall, this study goals to provide an encyclopedic analysis of the different machine-learning models used in Optical Character Recognition. The findings of this research will provide useful insights into the performance of different models and their suitability for different OCR applications.

2. Literature review

In recent years Optical Character Recognition has attained significant attention due to its potential applications in various domains, such as document digitalization, data extraction, and text analysis. This review of existing literature seeks to offer a comprehensive summary of the key advancements, challenges, and trends in OCR research, highlighting the methodologies, techniques, and applications that have emerged in the field. The historical development of OCR started from the mid-20th century [6].

Jamshed et al. [7] conducted research on handwritten OCR by performing a comprehensive literature survey, which served to present the cutting-edge results and techniques on OCR and highlighted research gaps. Om et al. [8] worked on odia characters. In the work, we can see processing input data and checking what are the things happening there. Using formulas of accuracy, sensitivity, and precision they made tables and result was gained. OCR post-processing techniques are surveyed by Thi et al. [9]. The contribution of OCR lies in highlighting the significance of enhancing the quality of OCR output and described as popular evaluation metrics. Manjusha et al. [10] proposed a character recognition method that is based on SVD(Singular Value decomposition) and k-NN (k-Nearest Neighbor). A comparison study between Tesseract, Amazon Textract, and Google Document AI was conducted by Thomas et al. [11] in his work. The paper aimed to evaluate and speed of these tools in recognizing text from images or digitized documents and to check the time took to complete the OCR process. The conclusion was amazing. Google Document AI or Amazon Textract will give the best accuracy whereas faster processing time is done by Tesseract.

OCR with neural network and post-correction with finite methods are done in the work of Senka et al. [12]. It was aimed to explore the improvement of the accuracy of OCR and focused on using Neural Networks for character recognition. An examination of the use of the post-correction method to improve the accuracy was conducted and also searched about the performance of the images and also compared the results with other types of OCR. The work resulted as the combination of neural networks and post-correction with finite state methods improves the accuracy of OCR compared to traditional OCR. Mansoor et al. [13] made a tool for OCR researchers that can be used for ranking different OCR algorithms. Bieniecki et al. [14] dealt with preprocessing tools before text recognition, especially with digital camera images. Carrasco et al. [15] developed an open-source tool that calculates statistics of the mismatches of a reference text with the output of an OCR engine. An African Buffalo optimized decision tree algorithm is developed by Archana et al. [16] with the assistance of the intelligent and collective behavior of the African Buffalos.

Yan et al. [17] used mostly used algorithms to make decision trees and visualize tree structure. Kotsiantis [18] tried to discuss the issues with decision trees and recent research trends in this area. An introduction of the Python library NetComp is done by Peter et al. [19]. Corrente et al. [20] through a didactic instance implemented the application of the deck of cards method. A new approach in OCR post-correction which is good enough for both unsupervised and supervised training is adopted by Dong et al. [21]. Lund et al. [22], by using maximum entropy classification methods, attempted to improve OCR. A focus on the accuracy of OCR in the early printed books by making use of a collaboration of convolutional and LSTM networks is carried out by Wick et al. [23]. The classification using tools such as fuzzy multilayer perception, and fuzzy support vector machine is demonstrated in the work of Chaunduri et al. [24]. Muna et al. [25] worked on English, Arabic, and Devanagari characters and very well narrated the difficulties that had occurred during the development of OCR. The advantages of the Tesseract model OCR are investigated by Ikha et al. [26].

Taking Tesseract as a case study, Christian et al. [27] demonstrated the efficiency and effectiveness of the specific approach to train the OCR engines. An algorithm to align the layouts which is made by different OCR models is prepared by Tomovic et al. [28]. Jaewoo et al. [29] presented a good multi-lingual OCR system for scanned documents. To overcome the drawbacks, they adopted REINFORCE algorithms rather taking supervised ones. A review on OCR is being carried out by Shubham et al. [30]. What are the achievements and difficulties of OCR engines is the thing discussed in the work. Using Open CV and Tesseract Ocr, license plate recognition technology was developed by Rithik et al. [31]. The central objective of their work is to build a precise image-processing technique together with the correct recognition of alphanumeric characters. Quinlan [32] focussed on decision trees for machine learning and broadly described ID3. A very basic algorithm is considered and new research directions are mentioned in the work.

Leonardo et al. [33] worked on MDT(multivariate decision trees) and tried to give out the best decision trees. They worked on 37 MDT induction algorithms and did experimental studies with 19 of them in 57 databases. Shambavi et al. [34] made an effort to do digitization work in the medical field, especially in ECG (Electro Cardio Gram). They made use of MATLAB-based tools to convert scanned ECG records into digitized ECG signals. A different aspect of OCR was seen in the paper of Xudong et al. [35]. With the usage of deep convolutional neural networks, the initial implementation and effective application of MOCR(Micron Optical

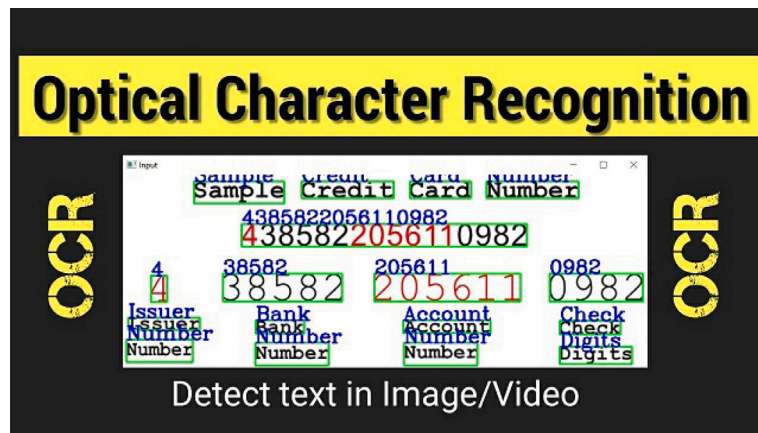


Fig. 1. Processing in an image by an OCR model.

Character Recognition) for making DFB(distributed feedback) laser chips are carried out. Khadija et al. [36] presented an instance of accessible OCR tools and They trained Tesseract on the Amazigh language, written in Latin characters.

Haoran et al. [37] put forward a model SDTR(Soft Decision Tree Regressor)which is a hierarchical differentiable neural regression model. It is just like a binary decision tree andwas assessed on various tabular regression tasks. Jakub et al. [38] analyzed data with the help of decision trees. A total of 804 training observations were taken which is from 18 youth soccer players and with the decision tree model the interpretation, and possibility of visualization are done in a well interesting manner. A budgeted Influence Maximization Problem or BIM Problem is being focused on by Suman et al. [39]. The proposed methodology had four steps, community detection, budget distribution, seed selection, and budget transfer. A maximization problem especially on conjugate duality is discussed in the work of Phan et al. [40]. They obtained zero gap duality for a vector-maximization problem.

3. Basic terminologies

3.1. OCR (Optical Character Recognition)

This is a technology which enables the detection and recognition of text from photos, and other digital sources. It is used to convert printed documents into editable documents which can be used more easily (see Fig. 1).

3.2. Amazon textract

Amazon Textract is a cloud-based OCR service offered by Amazon Web Services(AWS). It uses machine learning to automatically extract text and data from scanned images, PDFs, and other types of documents. This technology is designed to streamline document processing workflows and enable businesses to automate information extraction to improve efficiency and accuracy.

The primary use of Amazon Textract is to convert scanned documents and images into machine-readable formats such as text, structured data, or searchable PDFs. It supports character recognition for different fonts, tables, handwriting, and graphics, including barcodes, checkboxes, and other text shapes. Amazon Textract also offers features such as document analysis and form recognition that can identify and extract specific information from various types of forms, including W-9s, tax returns, invoices, and receipts. It can even recognize and extract data from unstructured documents such as contracts, resumes, and letters. Amazon Textract is easy to set up and use, with a simple API that can be integrated with different programming languages such as Python, Java, and NET. It is a scalable solution that can handle large volumes of data and provides real-time processing results.

3.3. Google document AI

Google Document AI is a suite of artificial intelligence tools for processing unstructured data in documents, such as text, tables, and images. It uses natural language processing and computer vision techniques to automatically extract information from documents. It is a technology developed by Google and it helps businesses save time and increase productivity by extracting data such as names, addresses, dates, and other key information which can be used for further processing and analysis. The technology is based on Google's expertise in natural language processing, and computer vision, and can help businesses improve their operations by making document processing faster and more accurate.

3.4. SVM

Support Vector Machine is a supervised ML program used for categorization and regression analysis. It works by finding a subspace whose dimension is one less than that of its ambient space (hyperplane) that separates the particulars into different grades. There are also variations of SVM such as Soft Margin SVM, Kernel SVM, and Multi-class SVM. SVM has been widely used in various domains, including image classification, text classification, bio-informatics, and finance, among others. It has a theoretical foundation and can handle high-dimensional data effectively.

3.5. Naive Bayes

This is a probabilistic ML algorithm that is as for classification. It stems from Bayes' theorem and works by calculating the probability of a given feature being associated with a certain class. It is used for classification tasks, particularly in natural language processing(NLP) and text classification. The "naive " assumption in Naive Bayes refers to the assumption of independence among the features, meaning the presence or absence of other features. It works as Training the labeled dataset, calculating the new data with the help of Bayes' theorem, and then classifying the data with the highest posterior probability. It is particularly well-suited for word classing assignments, in the same manner as spam filtering, opinion mining, and document categorization.

3.6. *k*-NN (*k*-Nearest Neighbors)

This is also a supervised machine learning algorithm made for sorting and retrogression analysis. It works by assigning a new datum to a class based on the *k*-nearest neighbors in a feature space. It is a distribution-free process, meaning it does not make any supposition about the fundamental distributions of the data. *k*-NN is considered as an instance-based or lazy learning algorithm because it does not build a model during the training phase. Instead, it stores all the training instances and uses them for making predictions. There are four phases of its working namely training(in this *k*-NN only memorizes the feature vectors), Prediction(classification-the algorithm selects the *k* nearest neighbor based on the distance), Voting(The class that appears most frequently among the neighbors is chosen), and Prediction(It is based on the majority class among the *k* nearest neighbors. In *k*-NN regression, the output values of the *k* nearest neighbors are averaged or weighted to make predictions.).

3.7. CNN (Convolutional Neural Network)

CNN is a deep-learning algorithm used for image classification and recognition. It can detect and identify image features in a pecking order. CNNs have revolutionized the scope of machine vision and have carried off state-of-the-art performance in image classification, object detection, image segmentation, and more. The important components of CNN are convolutional layers, stacking layers, dense layers, and activation functions. Their effectiveness in extracting meaningful features from visual data has made them an indispensable tool in the field of deep learning.

3.8. RNN (Recurrent Neural Network)

RNN is a deep learning algorithm used for sequence modeling and prediction. It can learn sequential relationships between input data to generate output predictions. It is designed to process sequential and temporal data, such as time series, text, speech, and video. RNNs are particularly effective in capturing and modeling dependencies and patterns in sequential data due to their recurrent connections and memory-like properties. The key component of an RNN is the hidden state, which represents the network's memory. During each step of the sequence, the hidden state is modified by considering both the current input and the preceding hidden state.

3.9. MNIST (Modified National Institute of Standards and Technology)

MNIST is a large public database of handwritten digits that is commonly used as a benchmark dataset for image classification. It consists of a large collection of 28*28 grayscale images of handwritten digits from 0 to 9. The MNIST dataset aims to train a model that can accurately classify or recognize the handwritten digits based on the pixel intensities of the images.

3.10. Precision

Precision, also referred to as positive predictive value, evaluates the accuracy of the positive predictions made by the model. It is calculated as the ratio of true positive (TP) predictions to the total of true positive (TP) and false positive (FP) predictions.

$$Precision = \frac{TP}{(TP + FP)} \quad (1)$$

3.11. Recall

Recall, also called sensitivity or the true positive rate, assesses the model's ability to correctly identify positive instances. It is calculated as the ratio of true positive (TP) predictions to the total of true positive (TP) and false negative (FN) predictions.

$$Recall = \frac{TP}{(TP + FN)} \quad (2)$$

3.12. F1 score

F1 score is a metric used to evaluate the performance of a machine learning model. It is the harmonic mean of precision and recall and is used as a balance between the two metrics.

The formula for finding the F1 Score is:

$$F1\ score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (3)$$

The span of this fraction is between 0 and 1, where 1 indicates perfect precision and recall, and 0 indicates the worst possible performance. The F1 score offers a balanced evaluation of a model's accuracy by taking into account both false positives and false negatives. It is especially valuable in cases where the dataset is imbalanced. The F1 score is calculated by considering both the model's ability to correctly identify positive instances (recall) and the precision of those identifications.

3.13. Accuracy

Accuracy refers to a performance metric that measures how well a model correctly predicts the outcome or class labels of a given dataset. The formula for calculating accuracy in machine learning is as follows:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (4)$$

$$\text{i.e., } Accuracy = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{False Positive} + \text{True Negative} + \text{False Negative}} \quad (5)$$

3.14. Machine learning models

Machine learning models are algorithms trained on a dataset to learn patterns and relationships between inputs and outputs. These models are used to make predictions on new, unseen data, and are widely used across various fields, including computer vision, healthcare, finance, natural language processing, and predictive analytics. There are various types of machine learning algorithms such as Supervised learning, Unsupervised learning, and Reinforcement learning.

3.15. Digital data

Digital data refers to any information that is stored in binary format, which means it is represented by a combination of 1s and 0s. It is data that is stored and processed using digital technology in opposition to analog data that is represented by continuous signals. It is typically stored in a computer or other digital devices. It encompasses a wide range of data types, including text, images, audio, video, and more. It is created, processed, and stored using electronic devices such as computers, smartphones, cameras, and various digital devices. It can be generated by humans, collected from sensors and instruments, or produced by automated systems. Digital data can be stored in various formats, such as documents (e.g. PDF, DOCX), images (e.g. JPEG, PNG), audio (e.g. MP3, AVI), databases, spreadsheets, and more.

3.16. Decision tree model

A decision tree model is a predictive analytics tool that utilizes a tree-like structure to represent decisions and their potential outcomes, including probabilities, costs, and benefits. It is a method for visualizing an algorithm that relies solely on conditional statements. In machine learning, a decision tree is a popular machine learning algorithm used for both classification and regression tasks. It builds a hierarchical structure of decision nodes and leaf nodes based on the features of the dataset. Each decision node tests a specific feature and based on the outcome, the algorithm moves down the tree to the next decision node or reaches a leaf node, which represents the final decision or prediction. Decision trees have several advantages such as their interpretability, as the decision-making process can be depicted and easily comprehended.

They can handle both numerical data, such as continuous variables and measurements, as well as categorical data, which includes discrete categories or labels. Additionally, they are robust to outliers, ensuring that extreme values do not significantly affect the performance of the model. Decision trees also require minimal data preparation, as they can handle missing values and do not require feature scaling. In the context of algorithms and libraries for decision tree modeling in Python, scikit-learn is a popular machine learning library that provides a decision tree implementation.

3.17. Fuzzy logic

Fuzzy logic is a method of reasoning and decision-making that addresses situations where uncertainty, vagueness, and ambiguity are present. Unlike traditional binary systems that rely on clear true or false values, fuzzy logic accommodates a range of truth values between 0 and 1. It is particularly useful when dealing with systems that have imprecise or uncertain inputs, making it well-suited for control systems, decision support, and artificial intelligence applications, where data may be inconsistent, unclear, or open to varying interpretations.

3.18. Membership function

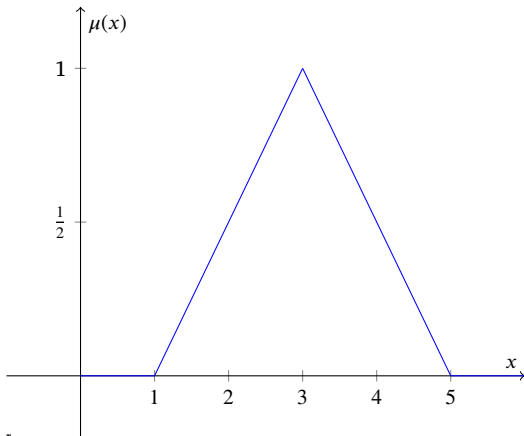
In fuzzy logic, a membership function is a mathematical function that defines the degree of membership of an element in a fuzzy set. Fuzzy sets allow for the representation of uncertainty and vagueness by assigning a membership value between 0 and 1 to each element in the set, indicating the degree to which the element belongs to the set. The different types of membership functions commonly used in fuzzy logic are as follows:

3.18.1. Triangular membership function

A triangular membership function is a type of fuzzy set membership function that is often used in fuzzy logic and fuzzy systems. The triangular membership function is characterized by three parameters: 'a', 'b', and 'c' where 'a' represents the starting point, 'b' represents the peak or center, and 'c' represents the ending point of the triangular function. The mathematical representation of a triangular membership function:

Given three parameters a , b , and c , the triangular membership function is defined as follows:

$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{b-a}, & \text{if } a < x \leq b \\ \frac{c-x}{c-b}, & \text{if } b < x < c \\ 0, & \text{if } x \geq c \end{cases} \quad (6)$$

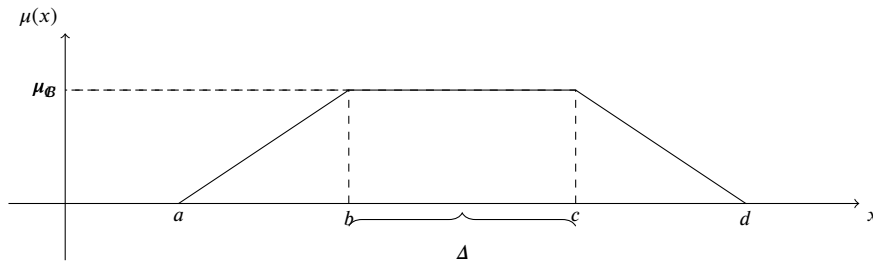


3.18.2. Trapezoidal membership function

A trapezoidal membership function is a type of fuzzy set membership function used in fuzzy logic and fuzzy systems. It is characterized by having four parameters that define the shape of the trapezoid. The mathematical representation of the trapezoidal membership function is:

The trapezoidal membership function is defined as:

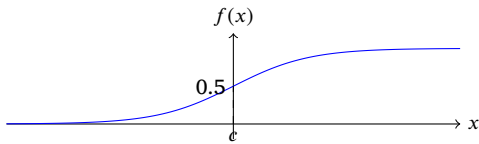
$$\mu(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x < b \\ 1 & \text{if } b \leq x \leq c \\ \frac{d-x}{d-c} & \text{if } c < x \leq d \\ 0 & \text{if } x > d \end{cases} \quad (7)$$



3.18.3. S-Shape membership function

An S-shaped (sigmoid) membership function is a type of membership function that the shape of an “S”. It is commonly used in fuzzy logic due to its smooth and gradual transition from 0 to 1 (or vice versa) around a specific midpoint.

$$f(x) = \frac{1}{1 + e^{-a \cdot (x-c)}} \quad (8)$$

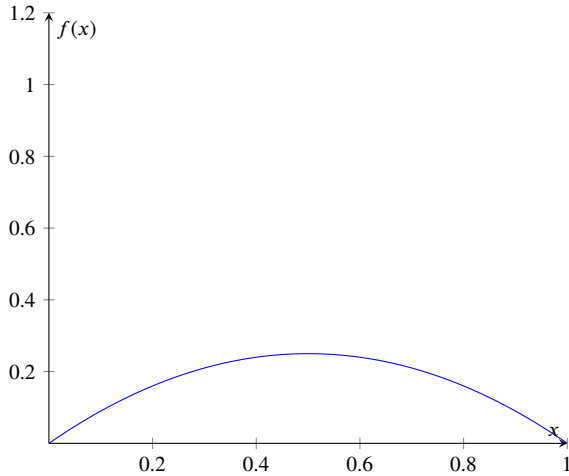


3.18.4. Beta-shaped membership function

The beta-shaped membership function is a fuzzy membership function, which takes the shape of a Beta distribution and a continuous distribution defined on the interval [0,1]. It is characterized by two parameters, often denoted by ' α ' and ' β ', which determine its shape.

The Beta-shaped membership function is given by:

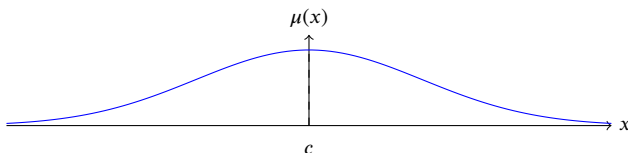
$$f(x) = x^{\alpha-1} \cdot (1-x)^{\beta-1} \quad (9)$$



3.18.5. Gaussian shape membership function

Gaussian shape membership function is a membership function, which takes the form of a Gaussian (bell-shaped) curve and is defined by two parameters: the mean (center) and the standard deviation (width) of the curve.

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right) \quad (10)$$



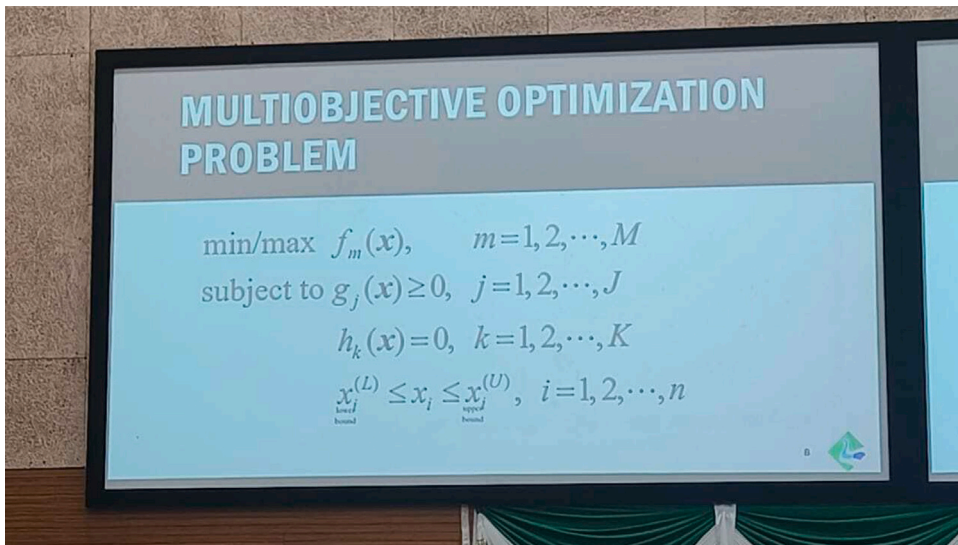


Fig. 2. Example of a figure we are taking for OCR.

4. Proposed system

Optical Character Recognition(OCR) system aims to recognize the printed text in an image which is written in English and also Mathematical equations. The work proposes to develop an OCR system that can be implemented using Tesseract, Easy OCR, Paddle OCR, MMOCR, Keras OCR, i2OCR and compare the most accurate model. The different machine learning algorithms we are implementing are Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). These are trained using a large corpus of labeled images and their corresponding text and help the machine to learn the patterns and features of different characters and fonts.

This dataset comprises 52 images captured with a mobile phone, encompassing a diverse array of real-world scenarios to rigorously test optical character recognition (OCR) systems. The images feature a variety of sources, including printed documents, computer monitors, projector screens, shopping receipts, medical reports, and banners. Each type presents unique OCR challenges, such as variations in font sizes, orientations, and text layouts. Additionally, the dataset includes both English text and mathematical expressions, providing a realistic assessment of an OCR system's capacity to handle different character sets and symbols. The images remain unadjusted and unprocessed to maintain authenticity, capturing common issues encountered in real-life image acquisition such as shadows, glare, and slight blurriness. This lack of preprocessing creates an opportunity to evaluate OCR systems under realistic conditions, where text may be partially obscured or displayed at various angles.

The first process is to consider a picture image that contains both Mathematical symbols and also English words. The next step is to put the picture into different OCR(Optical Character Recognition)models and try to compare their efficiency, and accuracy based on two attributes, one is the time taken to complete the task and the other one is recognizing the correct letters. We are taking some examples from real situations and applying them to different models. The pictures of preprocessing and post-processing are pointed out. The different tables are made from the calibration of different mathematical tools such as accuracy, and time taken in a wide variety of OCR models. The different stages are also narrated elaborately. Comparison with the aid of decision tree models and the finalization is done from that. Below is the figure we are considering for instance, (see Fig. 2).

The typical stages in the OCR process are

1. Preprocessing: Common preprocessing techniques include image scaling, rotation correction, noise reduction, and contrast enhancement.
2. Text Detection: The text detection algorithm aims to identify bounding boxes around text regions.
3. Text segmentation: If the image contains multiple lines of text, text segmentation separates the individual lines or words from each other.
4. Text recognition: In this, translation of the visual representation of characters into machine-readable text happens
5. Post-processing: It could involve language model-based corrections, spell-checking, and context-based analysis.
6. Output: This could be further processed for different applications, such as data entry, indexing, or searching (see Figs. 4 and 5).

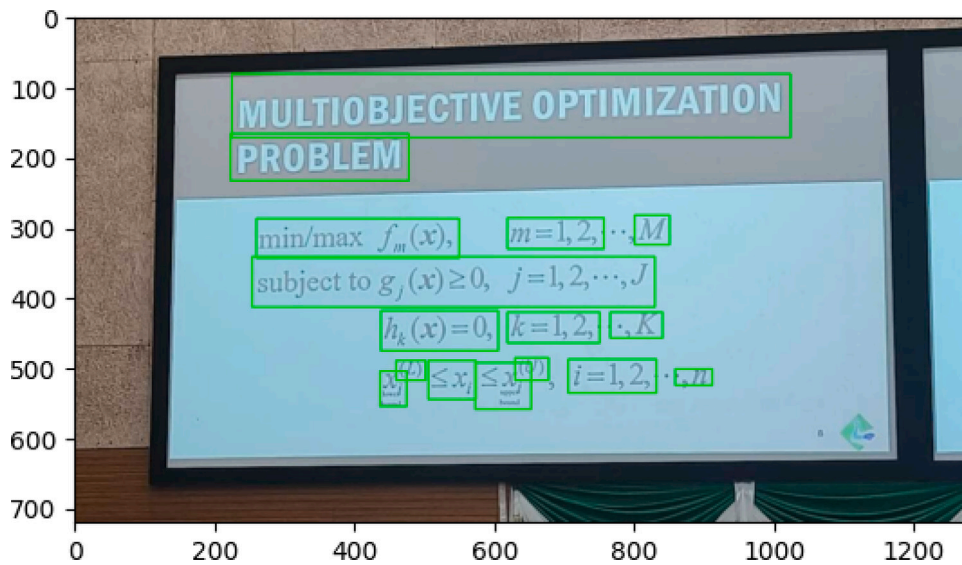


Fig. 3. Pre-processing.

	Text	Confidence
0	MULTIOBJECTIVE OPTIMIZATION	0.724624
1	PROBLEM	0.999626
2	min/max $f_m(x)$,	0.346095
3	$m=1, 2, \dots, M$	0.995760
4	M	0.999844
5	subject to $g_j(x) \geq 0, j=1, 2, \dots, J$	0.454137
6	$h_k(x) = 0, k=1, 2, \dots, K$	0.663874
7	$k=1, 2, \dots, K$	0.978212
8	(L)	0.975618
9	(U)	0.977990
10	$i=1, 2, \dots, n$	0.762529

Fig. 4. Post-Processing.

10	$i=1, 2, \dots, n$	0.762529
11	4	0.330124
12	42	0.396480
13	$\sim K$	0.619392
14	Sri	0.265171
15	,n	0.404036

Fig. 5. Post-Processing.

4.1. Easy OCR

Easy OCR is a free online OCR service that allows users to quickly and accurately convert digitized documents into editable text. It accommodates various image formats and is capable of detecting text in multiple Some common examples of easy OCR software include Adobe Acrobat, ABBYY FineReader, and Google Drive OCR. Table 1 shows the performance metrics of Easy OCR with the image shown in Fig. 3 (see Fig. 6).

Table 1
Performance metrics of Easy OCR.

Measures of efficiency	Algorithm	Value
Precision	$\frac{\text{Accurate Positive}}{\text{Accurate Positive} + \text{Inaccurate Positive}}$	0.75
Recall	$\frac{\text{Accurate Positive}}{\text{Accurate Positive} + \text{Inaccurate Negative}}$	0.75
F1-Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	0.75
Accuracy	$\frac{\text{Accurate Positive} + \text{Accurate Negative}}{\text{Total Samples}}$	0.6
Time taken	–	7 s
Equation	$(x - 0.75)^2 + (y - 0.6)^2$	–
Maximum value	–	1.1250

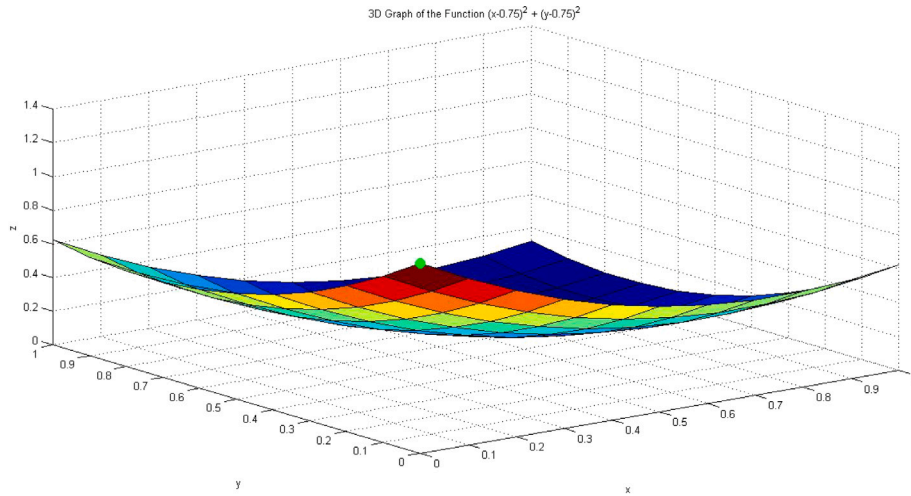


Fig. 6. The Maximization Graph of Easy OCR.

Table 2
Performance metrics of Tesseract OCR.

Measures of efficiency	Algorithm	Value
Precision	$\frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$	0.7142
Recall	$\frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$	0.7692
F1-Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	0.7406
Accuracy	$\frac{\text{True Positive} + \text{True Negative}}{\text{Total Samples}}$	0.6666
Time taken	–	7 s
Equation	$(x - 0.7142)^2 + (y - 0.6666)^2$	–
Maximum value	–	0.9544

The objective of the graph is to maximize $(x - 0.75)^2 + (y - 0.6)^2$. Max. the value obtained from the graph is 1.1250.

4.2. Tesseract

Tesseract is an open-source OCR engine developed by Google. It supports multiple languages and can handle various font styles and sizes. Table 2 shows the performance metrics of Tesseract with the image shown in Fig. 3 (see Fig. 7).

Aim of the graph is to maximize $(x - 0.7142)^2 + (y - 0.6666)^2$.

The optimum value from the graph is 0.9544.

4.3. Keras OCR

Keras is an open-source deep-learning library for building and training neural networks. It provides a high-level interface for building and running machine learning models. Keras is developed by using the Keras deep learning framework which supports

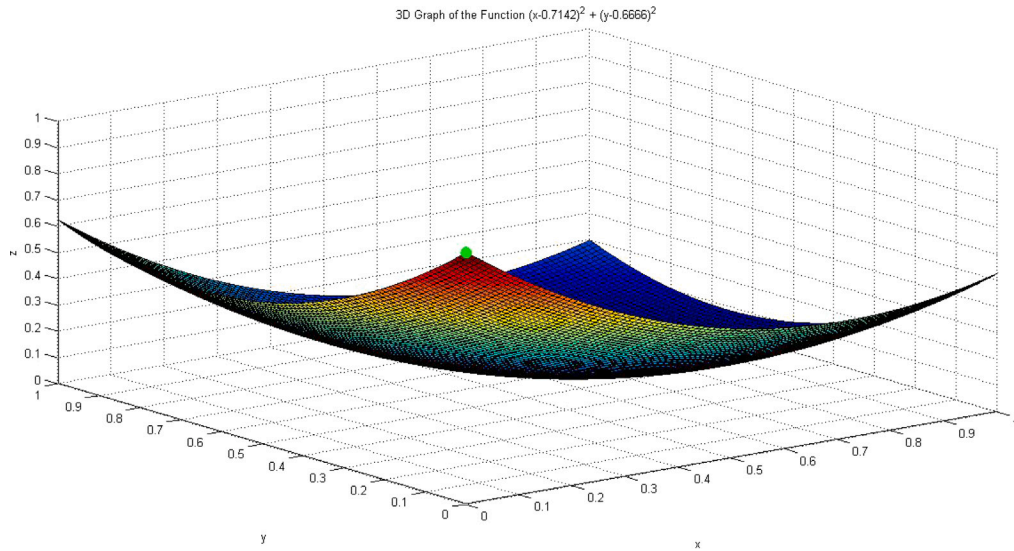


Fig. 7. The maximization graph of Tesseract.

Table 3
Performance metrics of Keras OCR.

Measures of efficiency	Algorithm	Value
Precision	$\frac{\text{Correct Positive}}{\text{Correct Positive} + \text{Incorrect Positive}}$	0.75
Recall	$\frac{\text{Correct Positive}}{\text{Correct Positive} + \text{Incorrect Negative}}$	0.60
F1-Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	0.6666
Accuracy	$\frac{\text{Correct Positive} + \text{Correct Negative}}{\text{Total Samples}}$	0.40
Time taken	–	20 s
Equation	$(x - 0.7142)^2 + (y - 0.6666)^2$	–
Maximum value	–	0.9225

languages, like English, Arabic, and Chinese. YOLOv3, CRNN, VGG16, CRAFT, TPS-ResNet-BiLSTM-Attn are the main neural networks used in keras OCR. Table 3 shows the performance metrics of Keras OCR with the image shown in Fig. 3 (see Fig. 8).

The objective of the graph is to maximize $(x - 0.75)^2 + (x - 0.4)^2$. Maximum value is 0.9225.

4.4. i2 OCR

i2OCR is an online OCR tool that converts scanned images and PDFs into editable text. It supports more than 60 languages and provides various output formats such as plain text, Microsoft Word, and searchable PDF. It extracts Math Equation text from images and scanned documents. We can edit, format, index, search and translate using i2OCR. Table 4 shows the performance metrics of i2OCR with the image shown in Fig. 3 (see Fig. 9).

The objective is to Maximize $(x - 0.4)^2 + (y - 0.2666)^2$. The maximum value from the graph is 0.8979.

4.5. Paddle OCR

Paddle OCR is a robust, flexible open-source Optical Character Recognition (OCR) offered by PaddlePaddle, a deep learning platform based on the PaddlePaddle library. It uses deep learning techniques to perform text detection, recognition, and layout analysis on images of text. It is designed to recognize complex texts in various languages and formats, including handwritten and printed text, digits, and special characters. Paddle OCR consists of deep learning platforms that support different neural networks to accurately recognize and convert images with text into editable digital formats, such as PDF, Word or Excel documents, or plain text files. Table 5 shows the performance metrics of Paddle OCR with the image shown in Fig. 3.

One of the key benefits of paddle OCR is its high accuracy. The OCR engine can recognize text with over 95% accuracy in various languages, including Chinese, Japanese, and Korean. Additionally, paddle OCR's deep learning algorithms enable it to handle

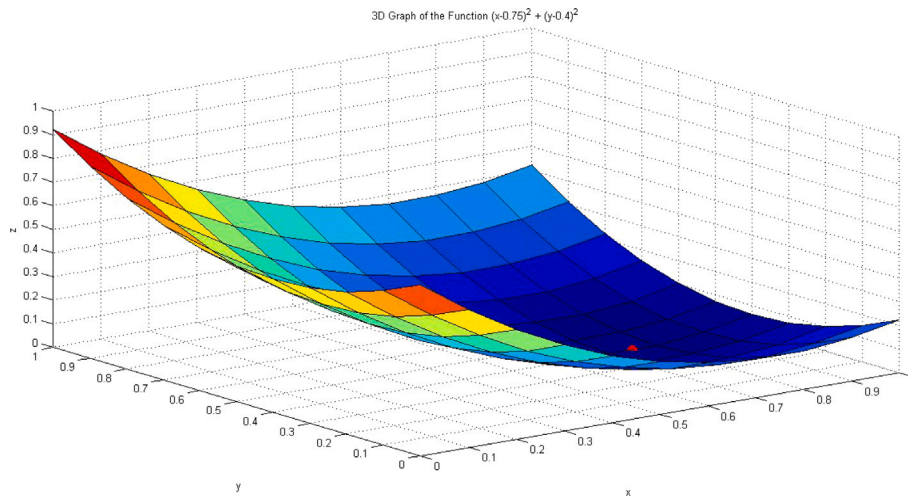


Fig. 8. The maximization graph of Keras OCR.

Table 4
Performance metrics of i2 OCR.

Measures of efficiency	Algorithm	Value
Precision	$\frac{\text{Natural Positive}}{\text{Natural Positive} + \text{Unnatural Positive}}$	0.4
Recall	$\frac{\text{Natural Positive}}{\text{Natural Positive} + \text{Unnatural Negative}}$	0.2857
F1-Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	0.3333
Accuracy	$\frac{\text{Natural Positive} + \text{Natural Negative}}{\text{Total Samples}}$	0.2666
Time taken	–	7 s
Equation	$(x - 0.4)^2 + (y - 0.2666)^2$	–
Maximum value	–	0.8979

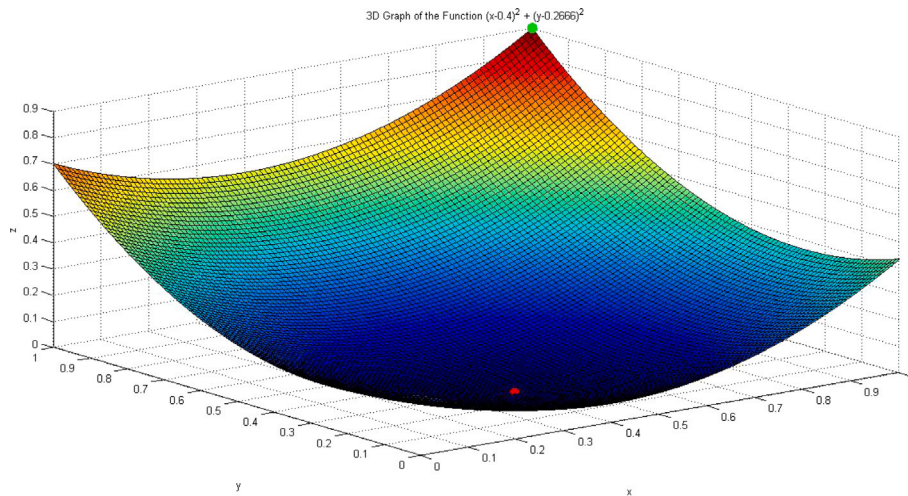


Fig. 9. The maximization graph of i2OCR.

complex text recognition tasks such as document image de-warping, noise cleaning, and image pre-processing. Paddle OCR is also designed to be user-friendly and easy to integrate with other applications. It comes with a user-friendly graphical interface(GUI) and provides simple APIs and SDKs for easy integration with other software tools. Overall, Paddle OCR is a powerful OCR engine that delivers high accuracy and advanced features for text recognition tasks. It is ideal for businesses and organizations that need to convert large volumes of complex text from paper documents and images into digital formats (see Fig. 10).

Table 5
The table of efficiency of Paddle OCR.

Measure of Efficiency	Algorithm	Value
Precision	$\frac{\text{pure positive}}{\text{pure positive} + \text{impure positive}}$	0.38
	$\frac{\text{pure positive}}{\text{pure positive} + \text{impure negative}}$	
Recall	$\frac{\text{pure positive}}{\text{pure positive} + \text{impure negative}}$	0.38
	$\frac{\text{pure positive}}{\text{pure positive} + \text{impure positive}}$	
F1 Score	$\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	0.38
Accuracy	$\frac{\text{Pure Positive} + \text{Pure Negative}}{\text{Pure Positive} + \text{Impure Positive} + \text{Pure Negative} + \text{Impure Negative}}$	0.333
	$\frac{\text{Pure Positive} + \text{Pure Negative}}{\text{Pure Positive} + \text{Impure Positive} + \text{Pure Negative} + \text{Impure Negative}}$	
Time taken	The processing time	30
Equation	$(x - 0.38)^2 + (y - 0.333)^2$	—
Maximum value	—	0.8293

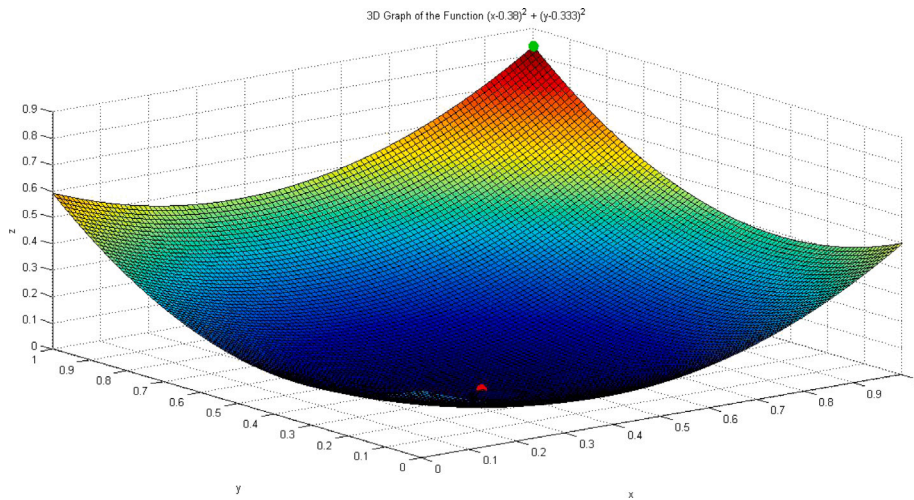


Fig. 10. The maximization graph of Paddle OCR.

Table 6
The table of efficiency of MMOCR.

Measure of Efficiency	Algorithm	Value
Precision	$\frac{\text{truly positive}}{\text{truly positive} + \text{falsely positive}}$	0.4444
	$\frac{\text{truly positive}}{\text{truly positive} + \text{falsely negative}}$	
Recall	$\frac{\text{truly positive}}{\text{truly positive} + \text{falsely negative}}$	0.3636
	$\frac{\text{truly positive}}{\text{truly positive} + \text{falsely positive}}$	
F1 Score	$\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	0.39996
Accuracy	$\frac{\text{truly positive} + \text{truly negative}}{\text{truly positive} + \text{falsely positive} + \text{truly negative} + \text{falsely negative}}$	0.2666
	$\frac{\text{truly positive} + \text{truly negative}}{\text{truly positive} + \text{falsely positive} + \text{truly negative} + \text{falsely negative}}$	
Time taken	The processing time	9
Equation	$(x - 0.444)^2 + (y - 0.2666)^2$	—
Maximum value	—	0.8466

The objective of the graph is to maximize $(x - 0.38)^2 + (y - 0.333)^2$. The maximum value is 0.8293.

4.6. MMOCR

MMOCR is a Python-based OCR library that uses machine learning to perform text recognition and extraction from images. It supports various OCR tasks such as text detection, recognition, and layout analysis. MMOCR (Multi Model Scene Text OCR) is an open-source OCR tool developed by the Multimedia Laboratory at the Chinese University of Hong Kong. It is constructed in such a way that it can recognize text in complex scenes, including varying fonts, complex backgrounds, and low contrast.

MMOCR provides a comprehensive collection of tools and models related to OCR tasks. Its key functionalities include text detection, text recognition, and text spotting in which text spotting is the combination of text detection and recognition tasks to detect and recognize text simultaneously. Table 6 shows the performance metrics of MMOCR with the image shown in Fig. 3 (see Fig. 11).

The objective of the graph is to maximize $(x - 0.444)^2 + (y - 0.2666)^2$. The maximum value is 0.8466.

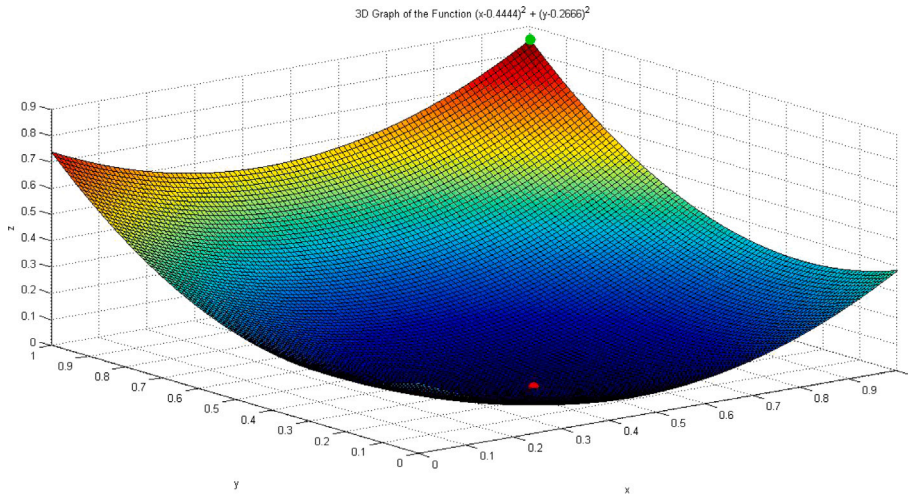


Fig. 11. The maximization graph of MMOCR.

Table 7

The table of efficiency of traditional OCR.

Measure of efficiency	Algorithm	Value
Precision	real positive	1
	real positive+unreal positive	
Recall	real positive	1
	real positive+unreal negative	
F1 Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	1
Accuracy	real positive+real negative	1
	real positive+unreal positive+real negative+unreal negative	
Time taken	The processing time	120
Equation	$(x-1)^2 + (y-1)^2$	–
Maximum value	–	2

4.7. Traditional OCR

Traditional OCR systems use rule-based or template-based techniques to recognize text characters in an image or document. It refers to the OCR technology and methods that were developed before the advent of deep learning and neural networks. The typical stages in a conventional OCR workflow encompass pre-processing, segmentation, feature extraction, classification, and post-processing, with each step being essential for transforming scanned images into text that can be read by machines. Table 7 shows the performance metrics of Traditional OCR with the image shown in Fig. 3 (see Fig. 12).

Objective is to maximize $(x-1)^2 + (y-1)^2$. The Maximum value is 2.

5. Comparison study

The comparison study of different OCR models with one image which includes both Mathematical and English words is represented. The content can be processed with line-wise or word-wise segmentation. Here the segmentation adopted is word-wise analysis. Proceedingly likewise, many more images can be taken for evaluation. Here the number of images considered is 52. From these training data, the comparison study is made among the models. For that, graphs of the respective models are plotted. Based on that, a decision tree for a more accurate conclusion is made. To select the most suitable OCR model for our task, we analyzed various image features and performance metrics. Image features such as resolution, size, word count, character count, and symbol count provided insights into image complexity, while performance metrics like accuracy, processing speed, and confidence scores helped evaluate the models' reliability and efficiency. By considering these factors, we aimed to identify the optimal OCR model for our specific use case. We constructed a decision tree model to train with these features. A comparative study of the performance of OCR models based on the speed and max value derived through the membership function is shown in Table 8. A total of 700 words were selected for analysis from 52 images, based on a confidence score threshold of 50%. The performance metrics of the different algorithms based on the decision tree analysis as shown in Table 9 (see Fig. 13).

6. Conclusion and future scope

Optical Character Recognition (OCR) technology converts images of text into machine-readable text, to make it easier for machines to read. There are many applications available with pre-trained models that are trained with deep learning using millions

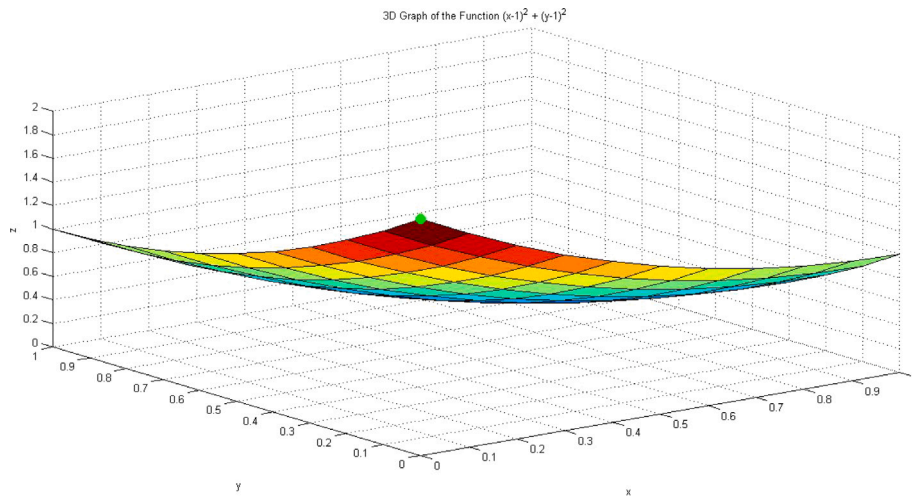


Fig. 12. The maximization graph of Traditional OCR.

OPTICAL CHARACTER RECOGNITION

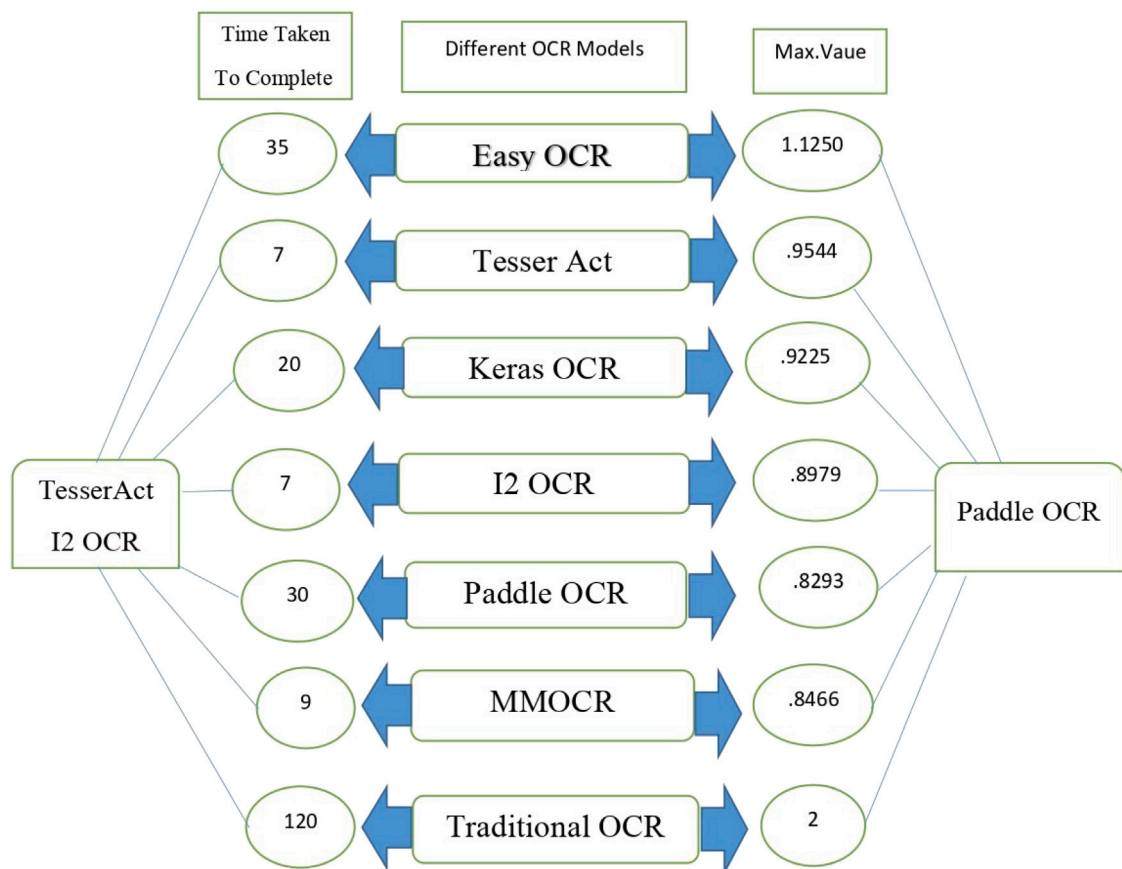


Fig. 13. The comparison table of OCR models.

Table 8
Performance metrics of OCR models.

Algorithm	Processing time (s)	Maximum value
Easy OCR	7	1.13
Tesseract	7	0.95
Keras OCR	20	0.92
i2OCR	7	0.90
Paddle OCR	30	0.83
MMOCR	9	0.85
TradOCR	120	2

Table 9
Performance metrics of OCR algorithms.

Algorithm	Accuracy	Precision	Recall	F1-Score
Easy OCR	0.6953	0.6931	0.7018	0.6942
Tesseract	0.6868	0.6847	0.6934	0.6858
Keras OCR	0.6784	0.6764	0.6850	0.6774
i2OCR	0.7037	0.7014	0.7103	0.7026
Paddle OCR	0.7136	0.7111	0.7201	0.7123
MMOCR	0.6643	0.6625	0.6709	0.6634
TradOCR	0.5982	0.5972	0.6048	0.5977

of pieces of data. The goal of this work is to find the best OCR models for mobile applications that create text from images. We employed different pre-trained models and datasets for analyzing image-to-text translation.

We conducted two different kinds of analysis in this paper. One is based on traditional machine learning, whereas the other is based on membership functions. Two decision variables were taken into consideration using the membership function analysis. One is the time, while the other is the highest value in each model function, which is regarded as the membership function with 0 and 2 as its extremes. The minimum of maximum values are figured out to get the best OCR model whose variable is less deviated from other taken values. At the same time, considering the time taken by each model, setting the minimum gives the fastest processing model. The two attributes, taken here specify the efficiency which is different in the two criteria.

In the analysis of traditional machine learning, the decision tree method employed in this research has proven to be a helpful tool for classification and prediction problems. By utilizing a tree-like structure of decision nodes and leaf nodes. The model was able to examine the features and ultimately assisting in reaching well-informed conclusions. Different image processing features are also utilized for this. Throughout the experiment the decision tree model demonstrated its ability to handle both categorical and numerical data, making it versatile for various types of datasets. It accurately identified the connection between the independent variables and the dependent variable, allowing for optimization. The pre-trained models all work well with our dataset, however, there are a few slight differences. According to our analysis, Tesseract and i2OCR are the best in terms of speed, whereas Paddle OCR is the most efficient in the prediction. Biasing may happen according to the variance in input data. In conclusion, this decision tree is proving that it is a valuable tool in this study, providing accurate predictions and classifications while maintaining interpretability. Its versatility, robustness, and ability to handle missing values make this decision tree a reliable option for various applications. Nonetheless, careful consideration should be given to avoid overfitting and enhance the model's generalization capabilities.

In this paper itself, the efficiency of different models has been checked which can be elaborated to explain the diagnostic field. The initiation of a medical-related algorithm has to be plotted in detail. Future possibilities in technology such as machine learning and deep learning can be included with the assistance of Optical Character Recognition in the prognosis area in the medical results. In future work, we propose several enhancements to improve OCR technology: incorporating advanced image pre-processing techniques, leveraging more sophisticated deep learning models, exploring hybrid approaches that combine different algorithms, utilizing large-scale training datasets, integrating contextual information, and implementing refined post-processing methods. These advancements have the potential to significantly boost the accuracy and reliability of OCR systems, making them more robust and effective for a wide range of applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Mahesh B. Machine learning algorithms-a review. *Int J Sci Res (IJSR)* 2020;9(1):381–6, [Internet].
- [2] Isheawy NA, Hasan H. Optical character recognition (OCR) system. *IOSR J Comput Eng (IOSR- JCE)* 2015. e-ISSN. 2278-0661.
- [3] Singh TP, Gupta S, Garg M. Machine learning: A review on supervised classification algorithms and their applications to optical character recognition in indic scripts. *ECS Trans* 2022;107(1):6233.
- [4] Baldominos A, Saez Y, Isasi P. A survey of handwritten character recognition with mnist and emnist. *Appl Sci* 2019;9(15):3169.
- [5] Memon J, Sami M, Khan RA, Uddin M. Handwritten optical character recognition (OCR): A comprehensive systematic literature review (SLR). *IEEE Access* 2020;8:142642–68.
- [6] Mori S, Suen CY, Yamamoto K. Historical review of OCR research and development. *Proc IEEE* 1992;80(7):1029–58.
- [7] Memon Jamshed, Sami Maria, Khan Rizwan Ahmed, Uddin Mueen. Handwritten optical character recognition(OCR): A comprehensive systematic literature review(SLR). *IEEE Access* 2020;8:3012542.
- [8] Jena Om Prakash, Pradhan Sateesh, Biswal Pradyut Kumar, Nayak Sradhanjali. Recognition of printed odia characters and digits using optimized self-organizing map network. In: *UTC from IEEE xplore*. 2020.
- [9] Nguyen Thi Tuyet Hai, Jatowt Adam, Coustaty Mickael, Doucet Antoine. Survey of Post-OCR Processing Approaches, 2021;54(6):3453476, Article 124.
- [10] Manjusha K, Anand Kumar M, Soman KP. Twenty-third national conference on communications. NCC, 2017.
- [11] Hegghammer Thomas. OCR with tesseract, amazon textract, and google document AI:a benchmarking experiment. *J Comput Soc Sci* 2022;5:861–82.
- [12] Drobac Senka, Linden Krister. Optical character recognition with neural networks and post-correction with finite state methods. *Int J Doc Anal Recognit (IJДАР)* 2020;23:279–95.
- [13] Alkhazi Alghamdi Mansoor A, Ibrahim S, Teahan William J. Arabic OCR evaluation tool. In: 2016 7th international conference on computer science and information technology. CSIT, IEEE; 2016, p. 1–6.
- [14] Bieniecki W, Graboski S, Rozenberg W. Image preprocessing for improving OCR accuracy. In: 2007 international conference on perspective technologies and methods in MEMS design. IEEE; 2007, p. 75–80.
- [15] Carrasco RC. An open-source OCR evaluation tool. In: *Proceedings of the first international conference on digital access to textual cultural heritage*. 2014, p. 179–84.
- [16] Panhalkar Archana R, Doye Dharmal D. Optimization of decision trees using modified African buffalo algorithm. *J King Saud Univ.- Comput Inf. Sci* 2022;34(8, Part A):4763–72.
- [17] Song Yan Yan, Lu Ying, Shanghai Archives of Psychiatry. Decision tree methods:applications for classification and prediction. 2015, <http://dx.doi.org/10.11919/j.issn.1002-0829.215044>, ISSN: 10020829.
- [18] Kotsiantis SB, Artificial Intelligence Review. Decision trees:A recent overview. 2013, <http://dx.doi.org/10.1007/s10462-011-9272-4>, ISSN: 02692821.
- [19] Wills Peter, Meyer Francois G. Metrics for graph comparison: A practitioner's guide. *PLoS ONE* 2020. <http://dx.doi.org/10.1371/journal.pone.0228728>.
- [20] Corrente S, Figueira JR, Greco S. Pairwise comparison tables within the deck of cards method in multiple criteria decision aiding. *European J Oper Res* 2021. <http://dx.doi.org/10.1016/j.ejor.2020.09.036>.
- [21] Dong R, Smith DA. Multi-input attention for unspecified OCR correction. In: *Proceedings of the 56 th annual meeting of the association for computational linguistics*. 2017, p. 70–6.
- [22] Lund WB, Walker DD, Ringger EK. Progressive alignment and discriminative error correction for multiple OCR engines. In: 2011 international conference on document analysis and recognition. IEEE; 2011, p. 764–8.
- [23] Wick C, Reul C, Puppe F. Comparison of OCR accuracy on early printed books using the open source engines. *Calamari OCRopus.JLCL* 2018;33:79–96.
- [24] Chaunduri A, Mandaviya K, Badelia P, Ghosh SK. Optical character recognition systems. In: *Optical character recognition systems for different languages with soft computing*. Cham, Switzerland: Springer; 2017, p. 9–41.
- [25] Awel Muna Ahmed, Abidi Ali Imam. Review on optical character recognition. *Int Res J Eng Technol* 2019.
- [26] Lestari Ikha Novie Tri, Mulyana Dadang Iskandar. Implementation of OCR using tesseract in detecting character in quotes text images. *J Appl Eng Technol Sci* 2022. <http://dx.doi.org/10.373885/jaets.v4i1.905>.
- [27] Clausner Christian, Antonacopoulos Apostolos, Pletschacher Stefan. Efficient and effective OCR engine training. *Int J Doc Anal Recognit* 2020. <http://dx.doi.org/10.1007/s10032-019-00347-8>.
- [28] Tomovic S, Pavlovic K, Bajceta M. Aligning document layouts extracted with different OCR engines with clustering approach. *Egypt Inform J* 2021. <http://dx.doi.org/10.1016/j.eij.2020.12.004>.
- [29] Park Jaewoo, Lee Eunji, Cho Nam Ik. Multi-lingual optical character recognition system using the reinforcement learning of character segmenter. *IEEE Access* 2020. <http://dx.doi.org/10.1109/ACCESS.2020.3025769>.
- [30] Srivastava Shubham, Verma Ajay, Sharma Shekhar. Optical character recognition techniques:A review. In: 2022 IEEE international students' conference on electrical, electronics and computer science. SCEECs 2022, 2022, <http://dx.doi.org/10.1109/SCEECs54111.2022.9740911>.
- [31] Rithik B, Raghav G, Aravind HS. Licence plate recognition system using open-CV and tesseract OCR engine. *Int J Eng Res Comput Sci Eng* 2022. <http://dx.doi.org/10.36647/ijercse/09.09.art003>.
- [32] Quinlan JR. Induction of decision trees. *Mach Learn* 1986. <http://dx.doi.org/10.1023/A:1022643204877>.
- [33] Canete-Sifuentes Leonardo, Monroy Raul, Medina-Perez Miguel Angel. A review and experimental comparison of multivariate decision trees. *IEEE Access* 2021. <http://dx.doi.org/10.1109/ACCESS.2021.3102239>.
- [34] Ganesh Shambavi, Bhatti Pamela T, Tridandapani Srin. Combining optical character recognition with paper ECG digitization. *IEEE J Transl Eng Heal Med* 2021. <http://dx.doi.org/10.1109/JTEHM.2021.3083482>.
- [35] Wang Xudong, Li Yebin, Liu Yong. Intelligent micron optical character recognition of DFB chip using deep convolutional neural network. *IEEE Trans Instrum Meas* 2022. <http://dx.doi.org/10.1109/TIM.2022.3154831>.
- [36] Gajoui Khadija El, Allah Fadoua Ataa, Oumsis Mohammed. Training tesseract tool for amazigh ocr. In: *Recent researches in applied computer science:proceedings of the 15 th international conference on applied computer science*. 2015.
- [37] Luo Haoran, Cheng Fan, Yi Yuqi. SDTR: Soft decision tree regressor for tabular data. *IEEE Access* 2021. <http://dx.doi.org/10.1109/ACCESS.2021.3070575>.
- [38] Marynowicz Jakub, Lango Mateusz, Andrzejewski Marcin. Predicting ratings of perceived exertion in youth soccer using decision tree models. *Biol Sport* 2022. <http://dx.doi.org/10.5114/BIOLSPORT.2022.103723>.
- [39] Banerjee Suman, Jenamani Mamata, Pratihar Dilip Kumar. ComBIM: A community-based solution approach for the budgeted influence maximization problem. *Expert Syst Appl* 2019. <http://dx.doi.org/10.1016/j.eswa.2019.01.070>.
- [40] Thach Phan Thien, Thang Tran Van. Conjugate duality for vector-maximization problems. *J Math Anal Appl* 2011. <http://dx.doi.org/10.1016/j.jmaa.2010.10.041>.